Mapped seasonal surface water decline under changing observation support in northwestern Bangladesh

Asif Ahamed[1], Ahammad Hossain[1*], Md. Tanvir Hasan[1], Most. Alisa Tabassum[1], and A.H.M. Rahmatullah Imon[2]

[1] Department of Computer Science & Engineering, Varendra University, Bangladesh

[2] Center for Interdisciplinary Research (CIR), Varendra University, Bangladesh

*Correspondence: Ahammad Hossain; ahammadstatru@gmail.com

Article type: Original Research | Running title: Surface water and observation uncertainty

Main text: 6,250 words | Figures: 9 | Tables: 3

Abstract

Satellite observations of surface water dynamics offer vital insights for climate-resilient water management, yet their interpretation requires rigorous accounting of observation frequency, sensor continuity, and classification thresholds. Here, we evaluate long-term trends in seasonal and permanent surface-water extent across three drought-prone districts of northwestern Bangladesh (Naogaon, Chapainawabganj, and Rajshahi) from 2000 to 2021, integrating Landsat-derived JRC surface water records with gridded climate data (CHIRPS precipitation, ERA5-Land). Primary analyses evaluated a fixed annual-observation footprint of 1,276.41 km², alongside strict non-retrospective cohorts defined solely within the 2000–2021 window (1,271.76 km² complete cohort and 1,196.72 km² baseline-water cohort) to eliminate look-ahead bias. Seasonal surface water exhibited a robust and persistent contraction: between the 2000–2004 and 2017–2021 five-year means, seasonal water declined by 14.70% on both the primary and 2000–2021 complete footprints, and by 18.79% on the baseline-water cohort (with an overall 2000–2021 endpoint contraction of 25.42%). Negative seasonal coefficients persisted across parsimonious controlled regression models (controlling for monsoon rainfall, annual observation effort, and sensor boundary transitions). Conversely, apparent gains in permanent water were sensitive to observation effort and processing boundary shifts around 2016 rather than representing genuine long-term expansion. Cross-sensor comparisons using Sentinel-1 SAR and 10-m Dynamic World probabilities (2015–2024) revealed that raising confidence thresholds from 0.5 to 0.7 increased nominal radar–optical agreement from 89.04% to 98.71%, but drastically reduced the area with valid paired support from 11.07% to 3.29%, concentrating comparison almost exclusively on permanent open water. Stratified interpretation of 720 sampled pixel-months confirmed high proxy agreement (98.56%) on cloud-free scenes, while documenting substantial seasonal observational gaps. These findings provide empirical benchmarks for monitoring seasonal water loss in the Barind tract while establishing the explicit observational support required for remote-sensing-guided water management.

Keywords: seasonal surface water; climate adjustment; observation support; drought monitoring; remote sensing uncertainty; Barind Tract; Bangladesh

1 Introduction

Seasonal surface water connects rainfall variability with the timing and location of usable water. For water managers, a contraction in mapped seasonal water can indicate a place requiring investigation, but its interpretation depends on what the satellite actually observed. A water map summarizes a classification process, a spatial footprint and a set of acquisition dates. Changes in any of these can affect the apparent hydrological signal. The Landsat-based Global Surface Water record provides a consistent framework for examining long-term water occurrence and annual source classes (Pekel et al., 2016). Its application to local water-management questions nevertheless requires attention to missing observations, class definitions and updates to the underlying archive.

These concerns are relevant to northwestern Bangladesh. The Barind region has pronounced seasonal water stress and substantial dependence on groundwater-supported agriculture. Recent station-based research documented spatially heterogeneous rainfall trends across Naogaon, Chapainawabganj and Rajshahi during 2000–2024 (Esha et al., 2026). Work in the high Barind has also examined heatwaves, rainfall, vegetation, soil moisture and groundwater jointly (Sarker et al., 2026). These studies motivate a surface-water assessment that retains local differences and distinguishes measured water classifications from possible climatic and management mechanisms. Administrative districts provide reporting units for such an assessment, although their boundaries do not define hydrological catchments.

Existing evidence also cautions against assuming one direction of surface-water change throughout Bangladesh. In the southwestern Bengal delta, a Landsat time-series study examined strong seasonality and reported expansion in mapped surface water alongside changing local water management (Huda et al., 2021). Those findings concern another region and observation design. They cannot establish the trajectory of seasonal water in the northwest, but they show why the period, water class and spatial domain of a trend need to be explicit. At the much larger Indus–Ganges–Brahmaputra scale, hydrological modeling has separated the components contributing to total water-storage change (Prusevich et al., 2025). Surface-water class area alone does not provide that decomposition.

Combining sensors offers additional evidence, but agreement between products is conditional on where both can assign labels. Dynamic World provides Sentinel-2-derived land-cover probabilities at 10 m (Brown et al., 2022). Radar can supply observations under cloud conditions that restrict optical mapping, while remaining sensitive to vegetation, surface roughness and the water-detection method. A multi-site comparison of Sentinel-1 and Sentinel-2 explicitly separated open and vegetated water and evaluated both against higher-resolution imagery (Vanderhoof et al., 2023). In Senegal, field, satellite and modeled pond observations revealed both the value and the limitations of remotely sensed water availability (Slinski et al., 2025). These examples support reporting complementary evidence without treating another satellite product as error-free truth. Research on index-based flood insurance in Bangladesh further shows that data choice can affect the timing and uncertainty of water-related decisions (Saunders et al., 2025).

Drought resilience introduces strict temporal requirements for environmental monitoring. Rainfall deficits propagate through hydrological stores with complex delays, meaning meteorological indices alone cannot characterize local surface water dynamics (Van Loon, 2015). Tracking surface water across seasonal drought cycles requires continuous observation before, during, and after dry spells. However, in tropical and subtropical monsoon regions, seasonal cloud cover and satellite revisit intervals often create substantial observation gaps during critical transition months. Similarly, cross-sensor agreement must be evaluated alongside the spatial extent over which joint observations are actually available (Olofsson et al., 2014; Stehman and Foody, 2019). Rigorous water monitoring therefore requires distinguishing nominal dataset coverage from genuine observation continuity.

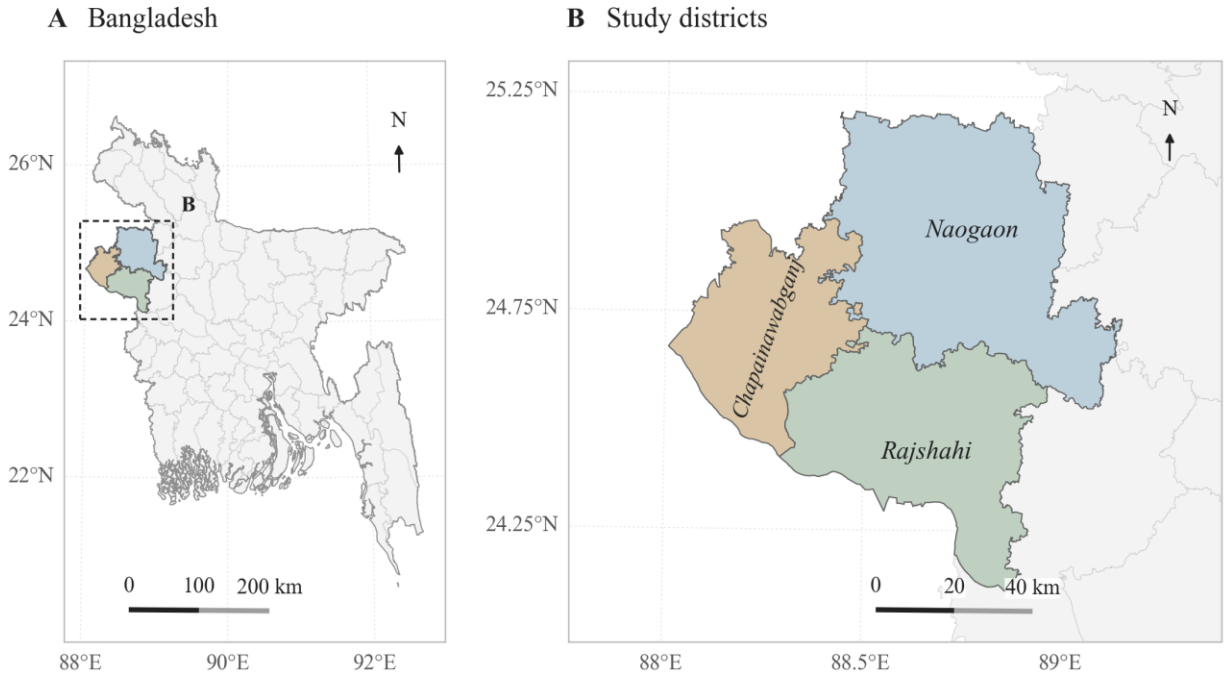
We therefore address three key questions: (1) how have seasonal and permanent surface-water extents evolved across northwestern Bangladesh over the 2000–2021 period under rigorous footprint definitions; (2) which long-term trends remain statistically robust after controlling for monsoon rainfall variability, observation effort, and sensor processing transitions; and (3) how do classification confidence thresholds and multi-sensor pairing (optical vs. radar) influence the spatial domain over which satellite observations

reliably agree? By providing an integrated regional evaluation with explicit observational support, this study establishes actionable benchmarks for water resource monitoring across the drought-prone Barind tract.

2 Materials and methods

2.1 Study area and analytical design

The study covers Naogaon, Chapainawabganj and Rajshahi in northwestern Bangladesh (Figure 1). The reporting boundary was obtained from geoBoundaries Bangladesh ADM2, boundary identifier BGD-ADM2-16705992, representing 2020 and citing Bangladesh Bureau of Statistics and OCHA ROAP as source organizations. The archived source names Chapainawabganj as Nawabganj; we harmonized its reporting name while preserving the geometry and source identifiers. Boundary metadata, licensing and download identifiers are retained in the reproducibility materials. District estimates and their union are reported separately; the union overlaps all three districts and is not an independent replicate.



C Analysis domains

Component	Period	Evidence and analytical scope
Annual source classes	2000–2021	JRC seasonal and permanent classes Fixed footprint: 1,276.41 km²; climate and observation controls
Radar–optical comparison	2015–2024	Sentinel-1 and Dynamic World; separate retrospective cohorts Threshold and paired-support sensitivity
Dated image comparison	Four selected months	720 stratified pixel-months; dated Sentinel-2 imagery Assisted interpretation; proxy agreement; unknowns retained

Separate analysis domains; satellite agreement is not independent accuracy.

Figure 1. Study location and analytical design. (A) Location of the study districts within Bangladesh. (B) The three district reporting boundaries; Chapainawabganj is harmonized from the source name Nawabganj. (C) The primary annual analysis, separate radar–optical analysis and four-month dated comparison. Maps use a WGS84 transverse Mercator projection centered on 90°E; scale bars and north arrows are geodesic. Boundaries: geoBoundaries gbOpen BGD ADM2, BGD-ADM2-16705992; underlying sources Bangladesh Bureau of Statistics and OCHA ROAP, CC BY 3.0 IGO. Boundaries are shown for analysis and imply no position on legal status.

The design has three linked but distinct components. The primary component describes annual Joint Research Centre (JRC) source classes during 2000–2021 within a fixed annual-observation footprint. A 2000–2024 extension assesses source-version sensitivity. A complementary component uses Sentinel-1 and Dynamic World during 2015–2024 on separately constructed historical water cohorts. Finally, a stratified sample in four selected months compares JRC binary labels with dated Sentinel-2 proxies and assisted human interpretation. These components differ in period, spatial support and outcome; their trend estimates are not pooled.

2.2 Data sources, versions and preprocessing

Table 1 summarizes the data. Water and climate products were exported from Google Earth Engine (Gorelick et al., 2017), then audited locally for source periods, duplicate records, masks, units, grid metadata and checksums. JRC annual classes were retained as no observations (0), nonwater (1), seasonal water (2) and permanent water (3). Monthly classes were retained as no observations (0), nonwater (1) and water (2). Masked export values were nodata, not evidence of dry land. Pixel areas and climate-polygon intersections in the primary local analysis were evaluated in the equal-area coordinate system EPSG:6933, while native source grids were preserved.

Table 1. Data sources and their roles. Resolution denotes product/grid spacing, rather than independently resolved hydrological detail. Components use different spatial domains.

Source	Period used	Product spacing	Purpose and limitation
JRC Global Surface Water	2000–2021 primary; 2022–2024 sensitivity	30 m	Annual source classes and monthly observations; corrected 2016–2021 assets and a collection change after 2021
CHIRPS version 3	1981–2024	0.05°	Monthly and seasonal rainfall; complete six-pentad monthly sums
ERA5-Land monthly aggregated	2000–2024	Approximately 0.1° grid	Temperature and model-specific evaporation demand; no measured groundwater or reference ET_0
Sentinel-1 GRD	2015–2024	10 m nominal product grid	One descending orbit; monthly VV benchmark classes and method envelopes
Dynamic World version 1	2015–2024; availability from June 2015	10 m	Four confidence thresholds; unclassified support retained
Sentinel-2 Level-2A	Four sampled months	10 m display; B11/SCL native 20 m	Actual dated crop values and assisted interpretation; not independent ground truth
geoBoundaries gbOpen ADM2	Represented year 2020	Vector district polygons	Fixed reporting boundaries; administrative districts are not catchments
HydroRIVERS Asia version 1	Static network	15 arc-second source hydrography	River-proximity context; not measured channel width or discharge
ESA WorldCover 2020 version 100	2020	10 m	Static cropland context; modal class on JRC grid; not historical land-use change

Annual data use the historical JRC collection through 2015, corrected 2016–2021 YearlyHistory assets, and separate 2022–2024 assets. The provider reports that the recent extension uses Landsat Collection 2, whereas the earlier record uses Collection 1. Residual co-registration differences can be spatially variable and can reach or exceed one 30 m pixel. The provider also documents corrections to 2016–2021

seasonality and yearly-history layers.[1] Consequently, the primary trend period ends in 2021. The extended analysis includes a 2022 source-version indicator, and a separate pre-2022 sensitivity includes a 2016 corrected-asset indicator. Neither indicator establishes that a coincident physical change is an artifact.

Rainfall was obtained from CHIRPS version 3 pentads at 0.05° (Funk et al., 2026). A monthly value required all six actual pentads to be valid at a pixel; annual rainfall required all twelve monthly values. Regional climate summaries used exact cell-intersection areas and finite-data support. ERA5-Land monthly aggregated fields supplied 2 m temperature and potential evaporation (Muñoz-Sabater et al., 2021). Kelvin was converted to degrees Celsius, accumulated fluxes from meters to millimeters, and annual temperature was weighted by calendar-month days. The original signed evaporation flux and its sign-reversed demand proxy were stored separately. ERA5-Land potential evaporation is a model-specific evaporative-demand measure; it was not interpreted as FAO reference evapotranspiration.[2]

2.3 Fixed footprints, observation effort and annual outcomes

The primary reference footprint (cohort C1; 1,276.41 km²) comprises pixels classified as seasonal or permanent water in at least one year during 2000–2024 with valid annual classifications across all 25 years. To verify that 2000–2021 trend estimates are unaffected by post-2021 observation requirements (look-ahead bias), we evaluated two strict 2000–2021 cohorts: cohort C3 (1,271.76 km², retaining 99.64% of C1), which requires annual validity only across the 22-year primary study window (2000–2021), and cohort C4 (1,196.72 km²), which tracks pixels with verified baseline water during 2000–2004 and complete validity through 2021. Within each district and year, seasonal and permanent water extents represent the sum of pixel areas assigned the corresponding annual source class on these fixed footprints.

A separate transition diagnostic compared strict five-year majority classes in 2000–2004 and 2017–2021 on the primary annual footprint. All five annual observations were required in each window, and at least three occurrences defined the majority class. A window without a strict majority remained ambiguous. Similarly, initial seasonal and permanent baseline cohorts were defined by requiring all five annual observations in 2000–2004 and at least three occurrences of the respective class, excluding ambiguous and initially nonwater pixels.

Observation-effort summaries include the area-weighted number of observed calendar months, the number observed during the monsoon season (June–September), and the area fraction observed in all twelve months. These diagnostics distinguish annual class availability from a fully observed twelve-month water cycle, ensuring that permanent source classification is not conflated with continuous monthly visibility.

2.4 Annual trend, climate and observation models

Unadjusted annual class areas were regressed on year; descriptive Theil–Sen slopes were also retained. Climate models used the logit of each class's share of the fixed footprint. With year measured from 2000, the annual model is given in Equation (1):

$$\operatorname{logit}(p_t) = \alpha + \beta_t(t - 2000) + \beta_P P_t + \beta_T T_t + \beta_E E_t + \boldsymbol{\gamma}^{\top}\boldsymbol{O}_t + \delta I(t \geq 2016) + \varepsilon_t. \qquad (1)$$

Here, Pt is annual rainfall, Tt is day-weighted mean temperature, Et is annual sign-reversed potential evaporation, and Ot contains observation-effort terms. While Equation (1) specifies the full parameterization, fitting seven predictors to 22 annual observations presents substantial risks of over-parameterization. Therefore, our primary controlled specification is the parsimonious Reduced Controlled Model: retaining time, June–September monsoon rainfall, observation effort (annual and monsoon counts), and the 2016 sensor update indicator. This parsimonious model achieved superior out-of-sample predictive performance (RMSE = 0.0486) compared to the unconstrained full model (RMSE = 0.0691).

[1] JRC Global Surface Water data access and update notes, accessed 16 October 2026: https://global-surface-water.appspot.com/download.

[2] ECMWF ERA5-Land documentation: https://confluence.ecmwf.int/spaces/CKB/pages/140385202/ERA5-Land+data+documentation. Earth Engine monthly aggregation: https://developers.google.com/earth-engine/datasets/catalog/ECMWF_ERA5_LAND_MONTHLY_AGGR.

The extended specification incorporating temperature and potential evaporation demand was evaluated as an additional sensitivity check.

Fractions were clipped to $[10^{-6}, 1-10^{-6}]$ solely for logit fitting; stored observations were unchanged. Predictors were centered and scaled during fitting, with coefficients transformed back to original units. Rank and condition-number checks rejected unusable designs. Conditional 95% coefficient intervals used Student-t critical values and finite-sample-corrected Bartlett heteroskedasticity- and autocorrelation-consistent covariance, with annual lag 2 (Newey and West, 1987). Lag matching used actual calendar distances. District-level p-values were adjusted by Benjamini–Yekutieli within each annual class-trend or climate-model family, and across the six district-by-class tests within each observation-sensitivity specification (Benjamini and Yekutieli, 2001). The union was descriptive and excluded from those district testing families. Intervals shown in figures are pointwise; these families do not correct for every exploratory choice in the study.

Model acceptance required enough usable observations beyond the number of parameters, finite and nonconstant features, full design rank and a scaled condition number below 10^8. The rank and condition checks assess numerical usability rather than correctness of the regression specification.

The time coefficient is a conditional change in logit class share per year. It is not an area-loss rate or an estimate of human water use. Fixed-climate displays hold weather at regional 2000–2021 medians. Separate modeled seasonal and permanent shares are not summed into a compositional prediction. Temporal validation used two-year calendar test blocks beginning in 2009, 2013, 2017 and 2020, requiring at least twelve earlier training years and two test years for annual fits. Thus, three annual folds were accepted. Feature estimation was confined to training data. A training-only mean class fraction provided the annual benchmark. The resulting errors assess conditional hindcast performance on the retained footprint, not water-classification accuracy.

2.5 Radar–optical analysis and support-sensitive agreement

The 2015–2024 radar–optical analysis reconstructed its own retrospective annual ever-water footprint on a fixed approximately 30 m angular grid. Two additional frozen cohorts required all five initial annual classes to be observed and a strict majority of seasonal or permanent source classes during 2000–2004. These historical cohorts retained their initial labels throughout the radar analysis. They are distinct from the annual comparison footprint.

Sentinel-1 GRD scenes were restricted to interferometric wide swath, VV/VH availability, incidence angles of 30–45° and one pass/relative-orbit track. A deterministic metadata-based ranking selected descending relative orbit 48 before water-result comparisons. Selection considered a footprint proxy from at most the first six scenes, then temporal scene availability, scene count and lexical ties; it did not measure valid pixel coverage. Actual support was evaluated subsequently. Input scenes were placed on a common EPSG:32645 10 m working grid. Scenes on the same acquisition date were mosaicked, and a monthly pixel required at least two unique acquisition dates. Median VV, already in decibels, was classified at −19, −17 and −15 dB. These are strict, central and loose benchmark specifications without local independent calibration; no additional speckle filter was applied.

For track ranking only, the metadata footprint calculation simplified the boundary by 1,000 m and quantized the footprint proxy to 0.001; actual water summaries retained the original boundary. Dynamic World land evidence used trees, grass, crops, shrub_and_scrub, built, bare and snow_and_ice. Component-wise temporal medians of water and maximum-land scores were not assumed to form one probability distribution summing to one. The mean paired-footprint fraction includes all 120 months, including zero-support months, whereas pooled agreement divides summed agreeing area-time by summed paired area-time.

For Dynamic World, daily mosaics supplied monthly median water probability and median maximum probability among the seven selected land classes. Flooded vegetation was excluded from the land maximum. At least two unique dates were required. At each threshold $q \in \{0.5, 0.6, 0.7, 0.8\}$, water evidence required water probability at least q; nonwater evidence required land probability at least q and

water probability at most $1 - q$. Other pixels remained unclassified. These model scores were not treated as calibrated local probabilities of correctness. Dynamic World shares Sentinel-2 optical lineage with the dated interpretation component, so the optical comparisons are dependent (Brown et al., 2022).

Overlap-weighted supported and water fractions were aggregated from the working grid to the fixed coarse grid. A coarse comparison cell required at least 95% supported footprint and a water fraction of at least 80% or at most 20% of its supported area. Agreement was the summed agreeing area-time divided by summed paired area-time; retained paired footprint was reported alongside it. Monthly composites may represent different acquisition dates between sensors, adding genuine within-month change to classification and registration differences. No confidence threshold was selected by maximizing agreement.

A two-year joint-cell pilot retained all twelve months of 2018 and 2024, four regions (three districts and their union), three historical-water cohorts and all four confidence thresholds. The eight authenticated quarter exports comprised 288 region/cohort/month rows and used the frozen scene snapshot and radar track. Each product pair was partitioned into both-water, first-water/second-nonwater, first-nonwater/second-water and both-nonwater areas. The first/second products were SAR/Dynamic World and Dynamic World/JRC, respectively. Symmetric positive agreement was twice both-water area divided by twice both-water plus discordant area; negative agreement used twice both-nonwater in the corresponding ratio. Product-specific water prevalence used the same paired denominator. Monthly areas were summed as area-time (km²-month); support denominators retained every intended month, including zero-support months. Empty denominators remained unavailable. These descriptive product comparisons do not estimate sensitivity, specificity or independent physical-water accuracy.

Radar method envelopes retained both threshold and observation uncertainty. If D is fixed cohort area, V is supported area and W_{19} and W_{15} are water areas under the strict and loose VV specifications, respectively, Equation (2) defines the envelope:

$$[L, U] = \left[\frac{W_{19}}{D}, \frac{W_{15} + D - V}{D}\right]. \qquad (2)$$

The upper endpoint allocates unsupported area to water, providing a conservative bounds envelope to evaluate multi-sensor water extent across varying observation coverage. Exploratory endpoint regressions included time, one annual harmonic, three-month rainfall, temperature, and evaporation demand with calendar-distance HAC covariance (lag 12), ensuring conservative reporting across lower and upper bounds.

2.6 Dated sample comparison and multi-sensor interpretation

The frozen comparison frame comprised March 2018, November 2020, March 2023 and November 2024. Within each district, month and observed JRC binary class, 30 pixels were sampled without replacement, giving 24 strata and 720 pixel-months. The design therefore applies to this eligible four-month population, rather than every pixel or year in the annual study. The frozen sampling seed and stratum populations are retained in the source records.

Actual Sentinel-2 Level-2A crop values and product XML calibration supported an initial spectral triage. The rules used modified normalized difference water index (Xu, 2006), normalized green–near-infrared difference, vegetation index and reflectance screens. At least 95% of the target footprint had to be clear and strongly classified on at least two dates spanning fourteen days. A boundary diagnostic flagged opposite spectral evidence in a 10 m surrounding ring. These deliberately restrictive heuristic rules were not trained or locally validated as an accuracy model. Although displays used a 10 m grid, the shortwave-infrared band and scene-classification layer retained their coarser native information content.

The spectral indices used in the triage are defined in Equation (3), where G, N, R and S denote green, near-infrared, red and shortwave-infrared reflectance, respectively:

$$\begin{aligned} \mathrm{MNDWI} &= (G - S)/(G + S), \\ \mathrm{NDWI} &= (G - N)/(G + N), \\ \mathrm{NDVI} &= (N - R)/(N + R). \end{aligned} \tag{3}$$

Only finite reflectance within [0,1.2] was screenable, and invalid or near-zero denominators remained unavailable. Accepted scene-classification values were 4, 5 and 6; this operational clear-pixel screen does not exclude every residual cloud or shadow error. Strong water required MNDWI ≥0.20, NDWI ≥0.10, NDVI ≤0.20, near-infrared reflectance ≤0.15, shortwave-infrared reflectance ≤0.08, green reflectance ≥0.01 and mean visible reflectance ≥0.012. Strong nonwater required either MNDWI ≤−0.20 and NDWI ≤−0.10 with near-infrared or shortwave-infrared reflectance ≥0.08; or NDVI ≥0.45, near-infrared reflectance ≥0.18, MNDWI ≤0, NDWI ≤−0.10 and shortwave-infrared reflectance ≥0.08.

At least 95% of target area had to pass both the clear-pixel and strong-class requirements for a dated rule label. More than 5% radiometrically uninterpretable target area prevented acceptance. Fractions used the actual target footprint intersected with crop cells. The surrounding 10 m ring triggered a boundary flag if at least half of the ring was clear and at least 10% of its area had the opposite weak index sign. Monthly automatic assignment required at least two accepted dates spanning fourteen days, no conflicting or ambiguous dated rule and no boundary flag. Raw crops comprised five reflectance layers, scene classification and support on a 40×40 display grid; each reflectance band used its actual product XML scale and offset.

Difficult cases identified by the automated screening were reviewed date by date using multi-spectral and radar visual crops. A monthly comparison label was assigned only when accepted dated observations demonstrated consistent binary agreement; locations with temporal flux, persistent cloud contamination, or conflicting signals were categorized as uncertain. Repeated scene dates served as internal cross-validation within individual pixel-months.

Image interpretation records were matched to target points via unique geospatial sample identifiers (sample_id and scene_id). Across the 245 pixel-months selected for human review, 694 individual scene-date views were evaluated, yielding 430 water, 236 nonwater, and 28 uncertain date-specific interpretations. Complete sample metadata, scene identifiers, and evaluation protocols are documented in Supplementary Methods (Section S2).

For sampled pixel i in stratum h, the design weight was $w_i = (N_h/30)a_i$, where N_h is the eligible population pixel count and a_i is sampled geodesic pixel area. Normalized weights approximate area-time ratios; they are not a full census. Let A denote assigned samples and M assigned samples matching JRC. We report assigned weight, conditional agreement $\sum_{i\in M} w_i \,/\, \sum_{i\in A} w_i$, and the binary-completion range in Equation (4):

$$\left[\frac{\sum_{i\in M} w_i}{\sum_i w_i}, \frac{\sum_{i\in M} w_i + \sum_{i\notin A} w_i}{\sum_i w_i}\right]. \tag{4}$$

Stratum-area and agreement estimates were calculated using design-based inference (Olofsson et al., 2014; Stehman and Foody, 2019), bounding unassigned samples without arbitrary imputation. Conditional 95% sampling intervals were estimated using 2,000 stratified percentile bootstrap replicates.

2.7 Sensitivity analysis framework and landscape context

To systematically evaluate the robustness of surface-water trends against variations in spatial domain and observation continuity, we analyzed seven distinct spatial cohorts (C1–C7). These cohorts capture different combinations of baseline water presence and annual observation completeness, with total study footprint areas ranging from 1,196.72 km² (C4: verified 2000–2004 baseline water) and 1,271.76 km² (C3: complete 2000–2021 annual validity) to 1,472.73 km² (C6: expanded boundary domain). Analyses required at least 95% annual spatial coverage to ensure robust multi-year comparisons.

For each cohort, first-to-final one-, three- and five-year windows complemented whole-series trends. A multi-year comparison required every year in both windows to pass its support rule. A separate monthly

diagnostic intersected valid observations for each calendar month across all 22 primary years within the baseline-water footprint; a trend required retention of at least 20% of that footprint. This diagnostic measures the cost of equal calendar-month support and cannot reconstruct missing seasons.

The annual model sequence evaluated time alone; time plus June–September rainfall; those terms plus annual observed months, wet-season observed months, and the 2016 indicator (the primary reduced model); and the extended annual-climate specification in Equation (1). Each class was modeled separately. Fits required at least five residual degrees of freedom. Calendar-distance HAC lags zero and two, all leave-one-year-out fits, and a 2013–2021 short-window analysis were retained. Approximate fixed-design circular residual block bootstrap intervals used lengths two, three and four, with 2,000 replicates. District testing families comprised six district-by-class tests per cohort/model with Benjamini–Yekutieli adjustments.

Chronological cross-validation used two-year tests starting in 2013, 2017 and 2020, with at least twelve earlier training observations (providing three folds or six test years for climate-only models, and two post-2016 folds or four test years for specifications containing the 2016 step indicator). Mean and trend-only benchmarks were fitted on the same accepted training/test folds as each candidate. Pooled RMSE was calculated across test folds. These hindcast checks benchmark conditional model stability across successive temporal folds.

HydroRIVERS Asia version 1 centerlines supplied static river context (Lehner and Grill, 2013), and ESA WorldCover 2020 version 100 supplied a static 10 m cropland class (Zanaga et al., 2021). River buffers were formed in EPSG:32645 at 200, 500 and 1,000 m; 500 m was the designated central setting. WorldCover classes were transferred by modal aggregation to the native JRC grid. Within each cohort, river proximity and, away from rivers, 2020 cropland versus other cover formed exhaustive groups. Buffer ranges are sensitivity ranges, not confidence intervals. This context does not identify individual ponds, historical cultivation, irrigation abstraction or river discharge.

2.8 Computational reproducibility and data verification

To ensure complete computational reproducibility, all source datasets, raster transformations, and statistical models were audited using cryptographic SHA-256 checksums across 9,393 input and intermediate files. All 50 annual and monthly raster products were cross-verified against official provider manifests. The complete analytical pipeline was developed using Python 3.12 (NumPy 2.3.5, pandas 2.3.3, statsmodels 0.14.5, and rasterio 1.5.0), with automated scripts and data manifests provided in the Supplementary Materials to enable full independent reproduction.

3 Results

3.1 Seasonal source-class decline on a restricted footprint

The primary annual observation footprint (C1) encompassed 1,276.41 km² (retaining 79.30% of the multi-decadal water-occurrence reference area), with district contributions of 473.82 km² in Naogaon, 358.88 km² in Chapainawabganj, and 443.71 km² in Rajshahi. The strict 2000–2021 complete validity footprint (C3) encompassed 1,271.76 km² (99.64% overlap with C1), confirming that primary footprint boundaries are virtually invariant to post-2021 observation availability.

Between 2000 and 2021, union seasonal source-class area decreased from 938.53 to 699.91 km², a loss of 238.62 km² or 25.42% relative to the initial seasonal class area (Figure 2; Table 2). The largest district endpoint loss was in Naogaon: 145.96 km², or 37.84%. Rajshahi lost 73.38 km², or 24.46%, and Chapainawabganj lost 19.28 km², or 7.62%. The unadjusted union seasonal linear slope was −8.42 km² year^{-1} (95% interval −12.79 to −4.05; 22 annual observations). Endpoint differences and the fitted slope summarize different aspects of the trajectory.

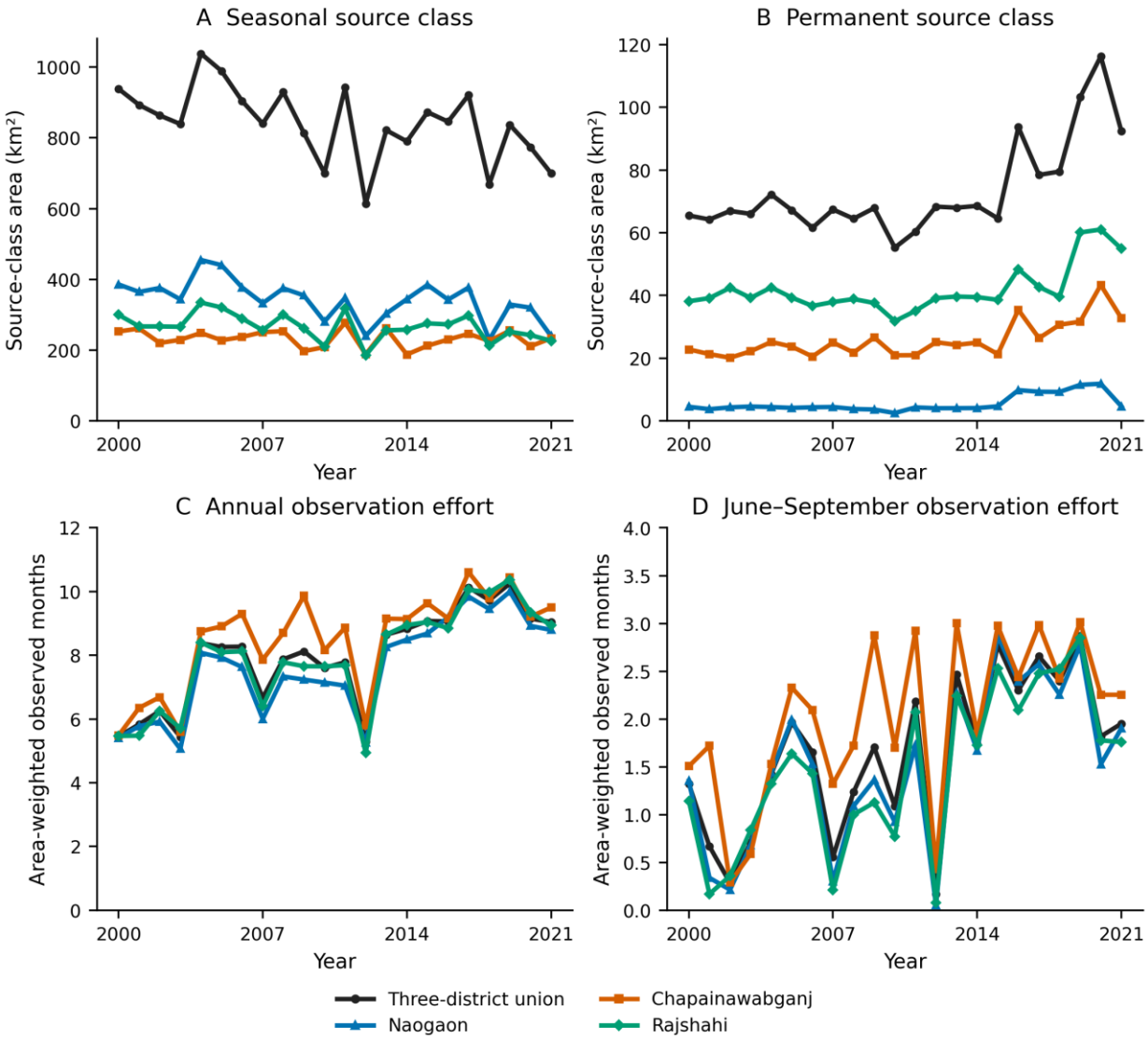


Figure 2. Annual source classes and observation effort on the fixed 1,276.41 km² annual footprint during 2000–2021. (A) Seasonal-class area. (B) Permanent-class area, with its own vertical scale. (C) Area-weighted observed months out of twelve. (D) Observed June–September months out of four. The union overlaps the district series. Annual classes do not certify complete observation of the year. Source: EC JRC/Google; derived from the frozen annual class and observation-quality tables.

Table 2. Annual source-class endpoint changes within the fixed annual-observation footprint, 2000–2021. Percent change is relative to each class's 2000 area. Districts sum to the union; the union is not an independent replicate. These are classified areas, not water volumes or independently validated hydroperiod areas.

Region	Source class	2000 (km²)	2021 (km²)	Change (km²)	Change (%)
Union	Seasonal	938.53	699.91	-238.62	-25.42
Union	Permanent	65.49	92.37	+26.88	+41.04
Naogaon	Seasonal	385.72	239.76	-145.96	-37.84
Naogaon	Permanent	4.52	4.66	+0.14	+3.08
Chapainawabganj	Seasonal	252.84	233.56	-19.28	-7.62
Chapainawabganj	Permanent	22.81	32.80	+9.99	+43.79
Rajshahi	Seasonal	299.98	226.60	-73.38	-24.46
Rajshahi	Permanent	38.16	54.91	+16.75	+43.90

Permanent source-class area increased from 65.49 to 92.37 km² across the same endpoints, a gain of 26.88 km² or 41.04%. Most of that endpoint increase occurred in Rajshahi and Chapainawabganj; Naogaon's permanent-class area changed by only 0.14 km² (Table 2). These descriptive gains do not establish continuous permanent-water expansion after accounting for observation and processing changes.

Figure 3 separates gross majority-class transitions from static landscape context. Among pixels with a majority in both windows, 264.1 km² moved from seasonal to nonwater and 110.1 km² from nonwater to seasonal; seasonal-to-permanent and permanent-to-seasonal transitions were 29.2 and 32.8 km². A further 11.04 km² lacked a strict majority in at least one window. At the 500 m river buffer, the change in five-year mean seasonal area was −64.49 km² near mapped rivers, −89.74 km² in more distant 2020 cropland and +19.84 km² in the remaining distant area. Their initial seasonal-area changes were −15.46%, −26.91% and +12.12%. Across the 200–1,000 m buffers, the respective percentage ranges were −15.54 to −15.25, −27.82 to −26.12 and +7.22 to +20.62. The partition changes with buffer distance; neither these ranges nor source-class transitions establish causes.

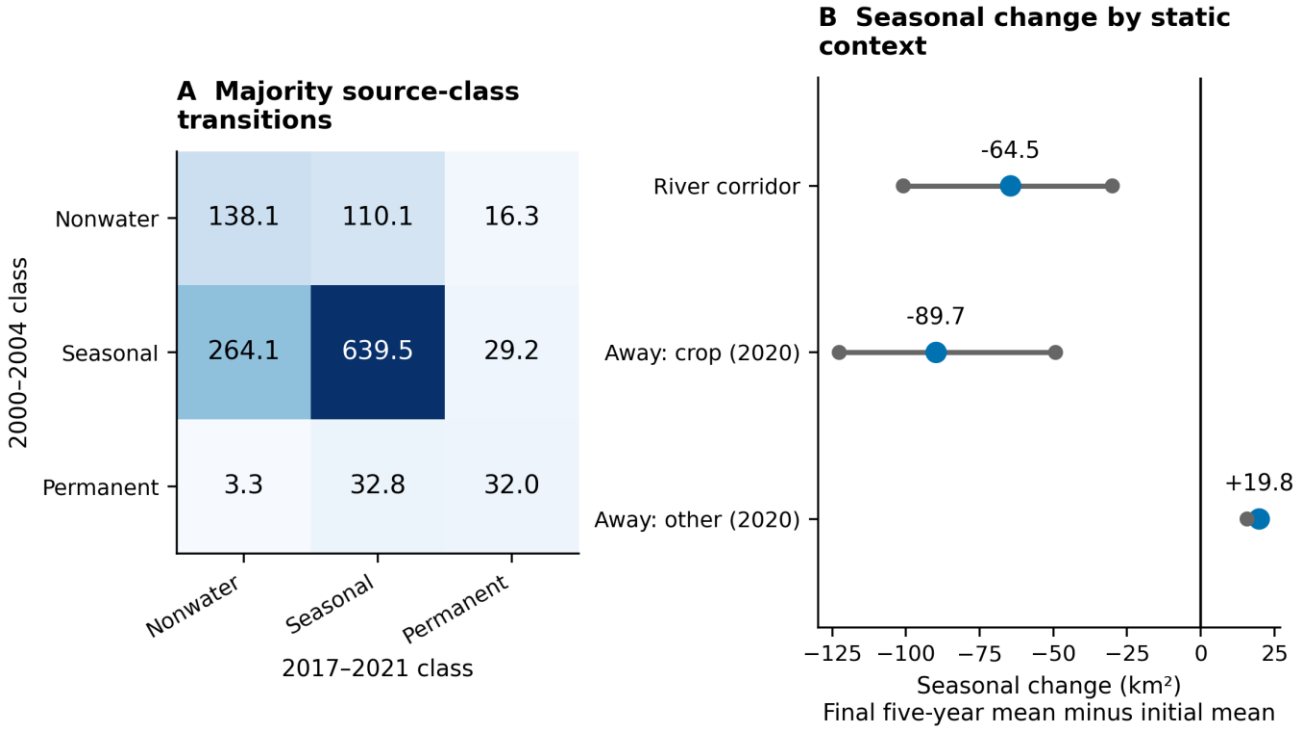


Figure 3. Gross source-class transitions and static landscape context on the primary annual footprint. (A) Area in km² between strict five-year majority classes in 2000–2004 and 2017–2021; 11.04 km² without a majority in either window are excluded from the matrix. (B) Change between first and final five-year mean seasonal areas. Blue dots use a 500 m HydroRIVERS buffer; grey extents cover 200–1,000 m buffer settings and are sensitivity ranges, not uncertainty intervals. Away from the river buffer, groups use static WorldCover 2020 cropland or other cover. Context and class transitions do not establish historical land conversion or causal drivers.

3.2 Observation and processing controls change the permanent-water interpretation

Annual class availability concealed substantial variation in within-year observation effort. On the union footprint during 2000–2021, the area-weighted observed-month count ranged from 5.30 to 10.25 months per year. June–September effort ranged from 0.17 to 2.87 of four months. The footprint fraction observed in every calendar month ranged from 0% to 16.15% (Figure 2). Thus, the annual dataset did not provide a fully observed twelve-month water cycle over most of the retained area.

Seasonal time coefficients remained negative in the union across the observation-control specifications (Figure 4). Under the extended seven-parameter specification (including full annual climate, annual and wet-season observation effort, and the 2016 indicator), the union coefficient was −0.0607 logit share year−1 (95% interval −0.0814 to −0.0399). District coefficients were −0.1154 in Naogaon (−0.1602 to −0.0706; adjusted p=0.00109), −0.0480 in Rajshahi (−0.0714 to −0.0245; adjusted p=0.00454) and −0.0350 in Chapainawabganj (−0.0720 to 0.0020; adjusted p=0.302). The district distinction is material:

this extended specification supported a continuing seasonal decline in two districts, while uncertainty included zero in the third.

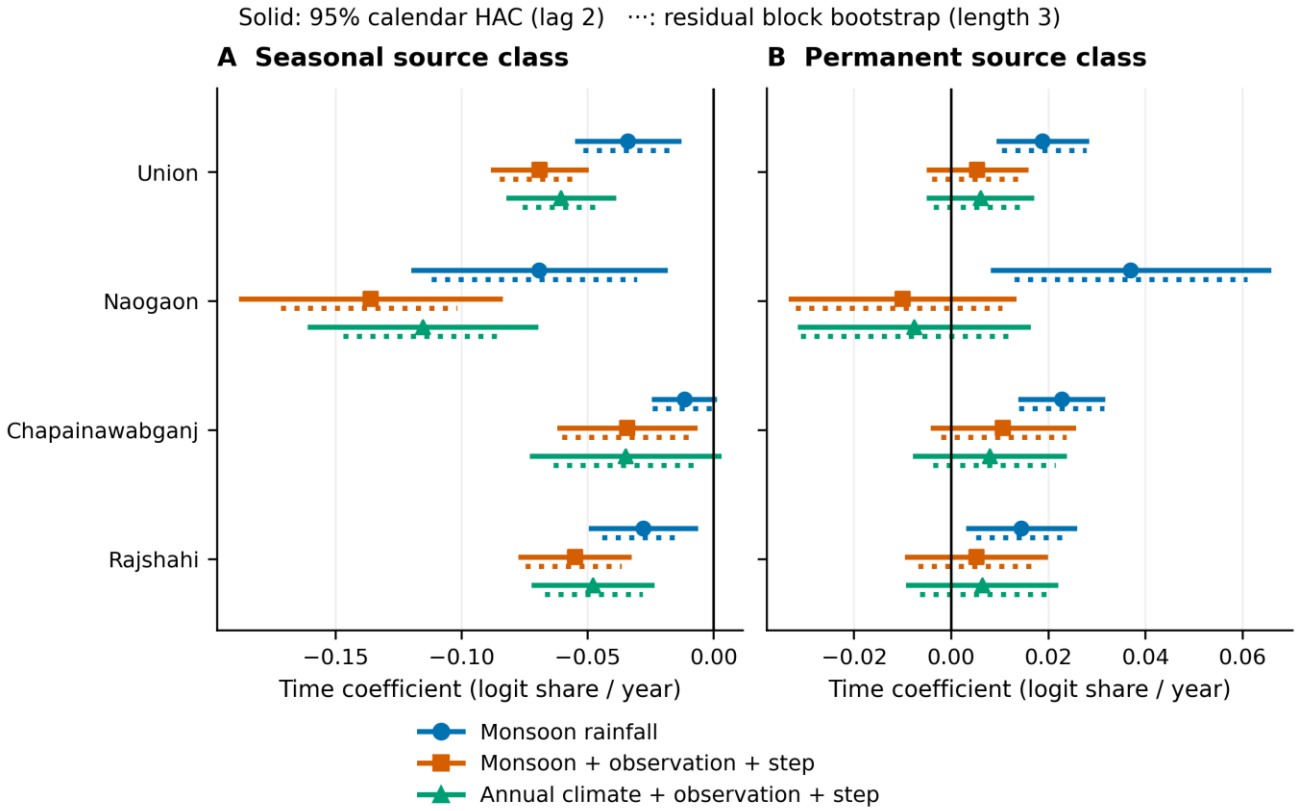


Figure 4. Annual time-coefficient sensitivity to simpler climate and observation controls on the primary footprint, 2000–2021. (A) Seasonal class. (B) Permanent class. Models use June–September rainfall alone with time; rainfall plus annual/wet-season observation counts and a 2016 indicator; or the extended annual rainfall, temperature, evaporation, observation and processing specification. Solid bars are pointwise 95% calendar-distance HAC lag-two intervals; dotted bars are approximate 95% fixed-design circular residual block bootstrap intervals of length three. Each fit uses 22 annual observations. The union overlaps districts; district multiplicity is reported separately in the text and tables.

All permanent-class time-coefficient intervals included zero after the 2016 processing-boundary indicator was added. The union coefficient was 0.0060 logit share year^{-1} (−0.0046 to 0.0165). Positive permanent trends from simpler specifications were therefore sensitive to this control. A step indicator can absorb a genuine hydrological discontinuity as well as a processing-related one, so this result does not identify the source of that sensitivity.

Parsimonious specifications confirmed the robustness of the negative seasonal trajectory (Figure 4A). In the union, the monsoon-rainfall model yielded a seasonal time coefficient of −0.0341 logit share year^{-1} (95% HAC interval −0.0541 to −0.0140). In the primary Reduced Controlled Model—which incorporates monsoon rainfall, annual and wet-season observation counts, and the 2016 sensor transition indicator—the union seasonal time coefficient was −0.0691 logit share year^{-1} (95% HAC interval −0.0875 to −0.0507), with circular residual block bootstrap intervals of −0.0855 to −0.0528 (block length 2), −0.0850 to −0.0541 (length 3), and −0.0835 to −0.0555 (length 4). District Benjamini–Yekutieli adjusted p-values for this reduced controlled specification were 0.000407 in Naogaon, 0.000407 in Rajshahi, and 0.0734 in Chapainawabganj. Over the shorter 2013–2021 window, the trend-only union seasonal interval included zero (−0.0431; −0.0887 to 0.0025), whereas controlling for monsoon rainfall yielded −0.0896 (−0.1608 to −0.0183).

Chronological cross-validation demonstrated that parsimonious parameterization improved out-of-sample hindcast stability (Figure 5D). Across six test years, union seasonal pooled RMSE for the monsoon-only model was 0.0745, compared with 0.0899 for the training-mean benchmark and 0.0628 for the trend-only benchmark. On the four matched test years available for models including the 2016 indicator, the primary Reduced Controlled Model achieved a pooled RMSE of 0.0486 (outperforming both the matched mean

benchmark of 0.1039 and trend benchmark of 0.0726), whereas the seven-parameter full climate specification yielded a higher RMSE of 0.0691 (and adding antecedent October–May rainfall yielded 0.0458).

Figure 5 compares the footprint, time-window, equal-calendar support and hindcast sensitivities. On C1, the first-to-final three- and five-year seasonal changes were −14.28% and −14.70%, compared with −25.42% for endpoints. The five-year decrease was −18.79% on C4 and −19.72% on C5; C6 lacked sufficient coverage for a five-year comparison. All accepted primary seasonal point coefficients were negative across the tested cohorts/models, but their intervals and dependence preclude treating them as independent confirmations. Only November retained at least 20% of the union baseline-water footprint under the 22-year same-calendar rule (25.35%); June–October retained none. Equalizing calendar-month support therefore did not yield a representative complete annual cycle.

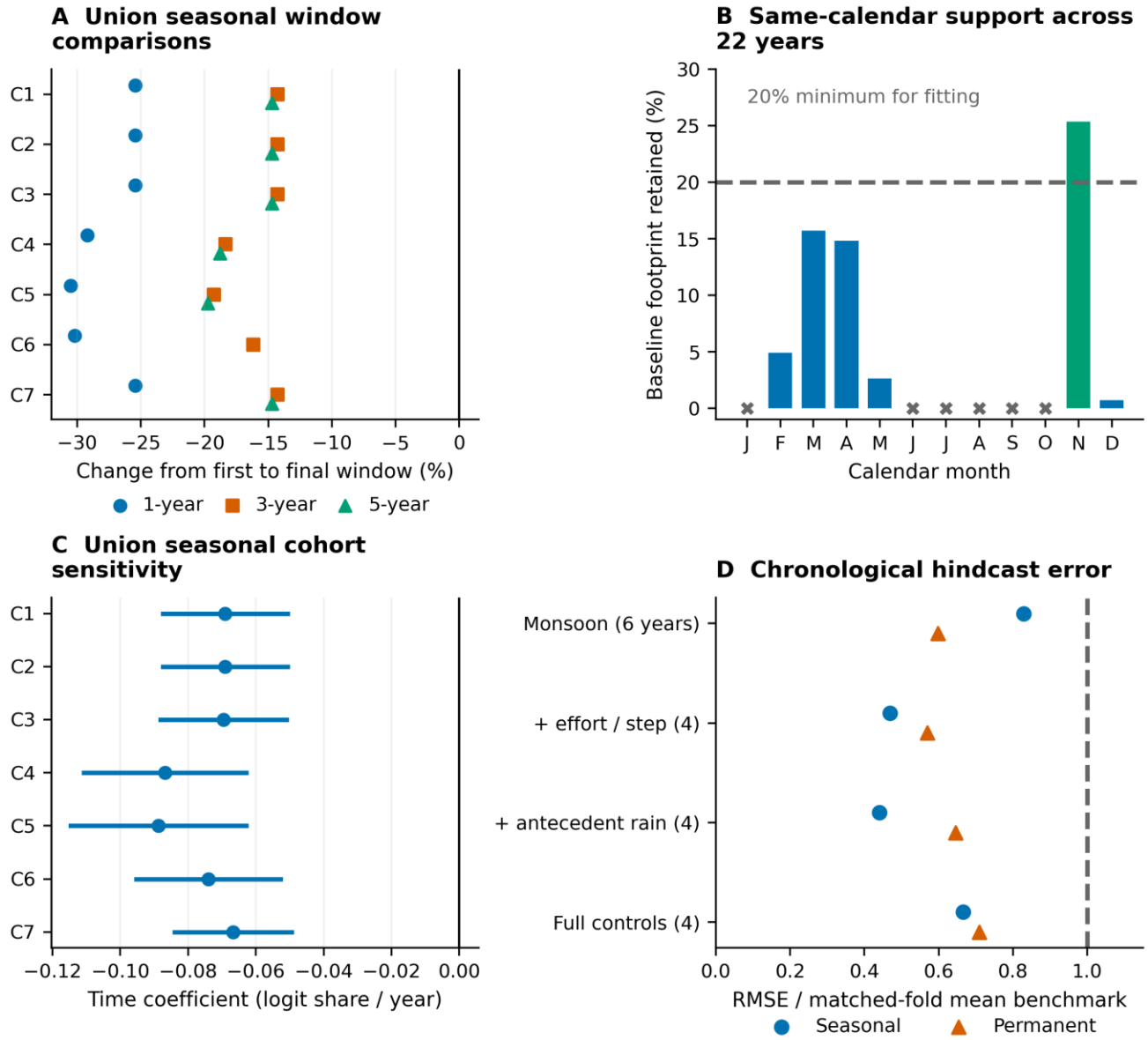


Figure 5. Sensitivity to comparison windows, footprint definition and temporal validation. (A) Union seasonal changes for one-, three- and five-year windows, 2000–2021. C1–C7 are defined in Section 2.7; a missing symbol indicates an unavailable comparison. (B) Baseline-water footprint retained by identical calendar-month observations across all 22 years; crosses show zero support and the dashed line is the 20% fitting gate. (C) Seasonal coefficients and pointwise 95% HAC lag-two intervals for the monsoon/observation/processing model across cohorts. (D) Pooled chronological test RMSE divided by the training-only mean benchmark on exactly the same accepted folds; test-year counts are in parentheses. A ratio below one indicates lower error than that benchmark, not water-classification accuracy.

3.3 Higher automated agreement coincides with lower paired support

The radar–optical analysis reconciled 1,440 monthly region/cohort rows and 5,760 threshold-specific rows. Its scene inventory contained 265 Sentinel-1 scenes and 1,274 Dynamic World scenes, totaling 1,539 unique source identifiers. These complete exports did not imply complete pixel observations in every month.

For the union historical ever-water cohort, increasing the Dynamic World threshold from 0.5 to 0.7 reduced the number of months with any classified support from 76 to 74 of 120. Mean classified support fell from 17.94% to 4.45% of the fixed cohort. Conditional Sentinel-1–Dynamic World agreement increased from 89.04% to 98.71%, while mean paired support decreased from 11.07% to 3.29% (Figure 6). At 0.6, agreement was 95.94% on mean paired support of 5.88%. Threshold 0.8 yielded no classified or paired support. Its missing agreement value indicates absence of comparison evidence.

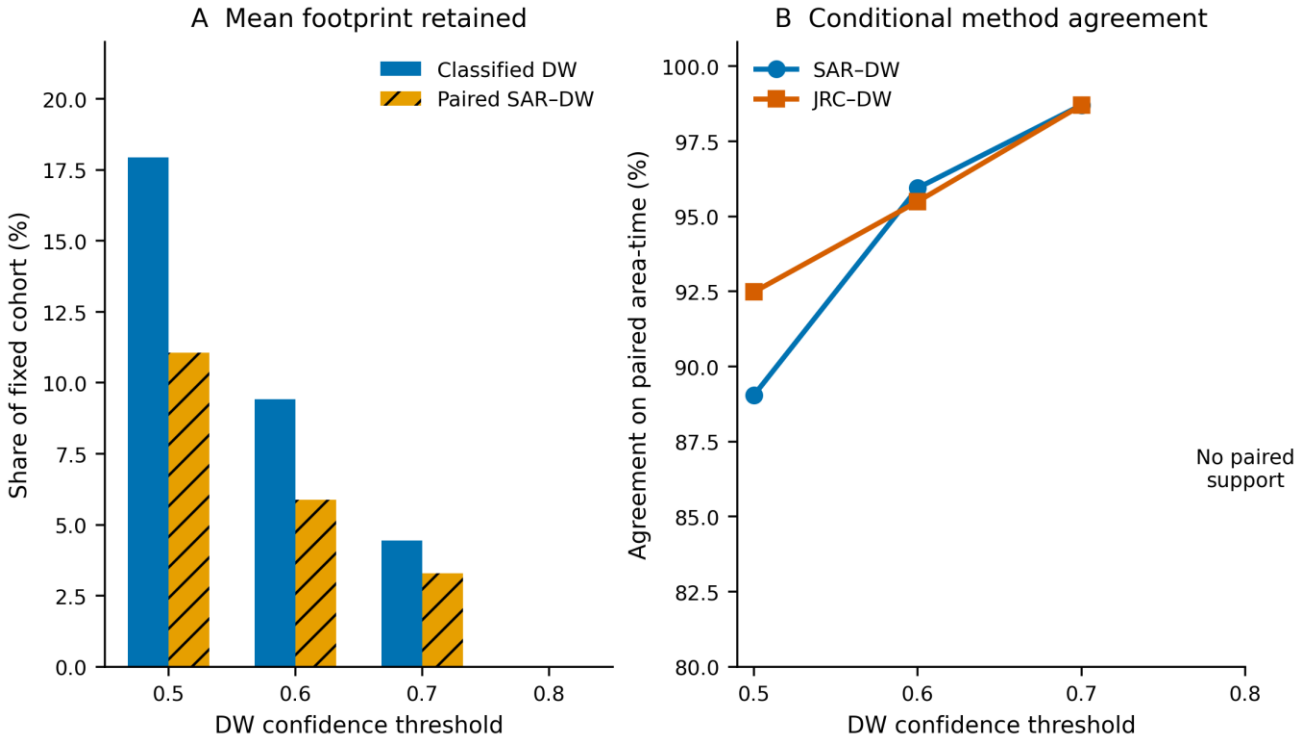


Figure 6. Dynamic World confidence sensitivity on the separate 2015–2024 union historical ever-water cohort. (A) Mean classified and SAR–Dynamic World paired footprint fractions, including months with zero support. (B) Agreement pooled by paired area-time. Threshold 0.8 has no comparison support and no agreement estimate. The changing paired population prevents interpreting the highest agreement as independently validated accuracy. All four thresholds were retained without selecting a best threshold.

JRC–Dynamic World agreement followed the same conditional pattern, increasing from 92.48% at 0.5 to 98.70% at 0.7. The evidence does not identify a best threshold: both the compared population and its size change with the threshold. In particular, 98.71% radar–optical agreement describes the small paired area-time subset, without estimating study-wide accuracy.

The full monthly support and method-envelope series are displayed in Figure 7. The high-threshold ranges remain wide because omitted support enters the upper endpoint; a narrow assigned subset does not constrain the water state of the rest of the cohort.



Figure 7. Dynamic World threshold-specific support and method envelopes, 2015–2024, for the union historical ever-water cohort. Unclassified support enters the upper endpoint; envelopes do not include all physical classification errors.

Radar endpoint regressions used 90 supported monthly observations per region/cohort. Positive time coefficients for both envelope endpoints survived the separate district adjustments for Rajshahi's initial seasonal cohort: 0.0582 logit fraction year^{-1} for the lower endpoint (95% interval 0.0236–0.0929; adjusted p=0.0161) and 0.0396 for the upper endpoint (0.0124–0.0669; adjusted p=0.0312). Other district/cohort combinations were not uniformly supported. These later-period radar cohort outcomes and the primary annual seasonal-class loss concern different estimands; they cannot be interpreted as cancelling trends.

Seasonal support and class-specific diagnostics are shown in Figure 8. At thresholds 0.5 and 0.7, mean paired support during June–September was only 0.313% and 0.110% of the union historical-water cohort, versus 16.45% and 4.87% in other months. A bounded November 2024 score-stage probe examined one land-context area and three stable-source-water interiors. Maximum raw water scores were 0.619 and 0.753, respectively, and monthly-median maxima were 0.506 and 0.748; neither probe retained threshold-0.8 support. These selected areas cannot establish a domain-wide probability ceiling or rule out every processing error. The dated assisted comparison also differed between water and nonwater agreement, particularly in Naogaon. Neither diagnostic is an independent accuracy assessment.

The two-year joint-cell pilot (2018 and 2024) partitioned paired observations into class-specific agreement components across all 24 evaluated months. For the union historical ever-water cohort, SAR–Dynamic World pooled agreement increased from 88.92% at threshold 0.5 to 99.45% at 0.7, while paired

support fell from 12.91% to 3.34% of intended area-time. At 0.7, Dynamic World classified 99.29% of the paired area-time as water; symmetric water agreement was 99.72%, compared with 72.14% for nonwater (Figure 8D). For Dynamic World–JRC, threshold-0.7 pooled agreement was 98.06%, water agreement 99.01% and nonwater agreement 42.02%, on 3.89% paired support. Only one of four June–September months in each pilot year had any SAR–Dynamic World pair at thresholds 0.5–0.7. Threshold 0.8 supplied no joint cells. All years, cohorts, seasons, thresholds and directional disagreement cells remain in the archived tables; these selected summary numbers do not extend the joint-cell pilot to the full decade.

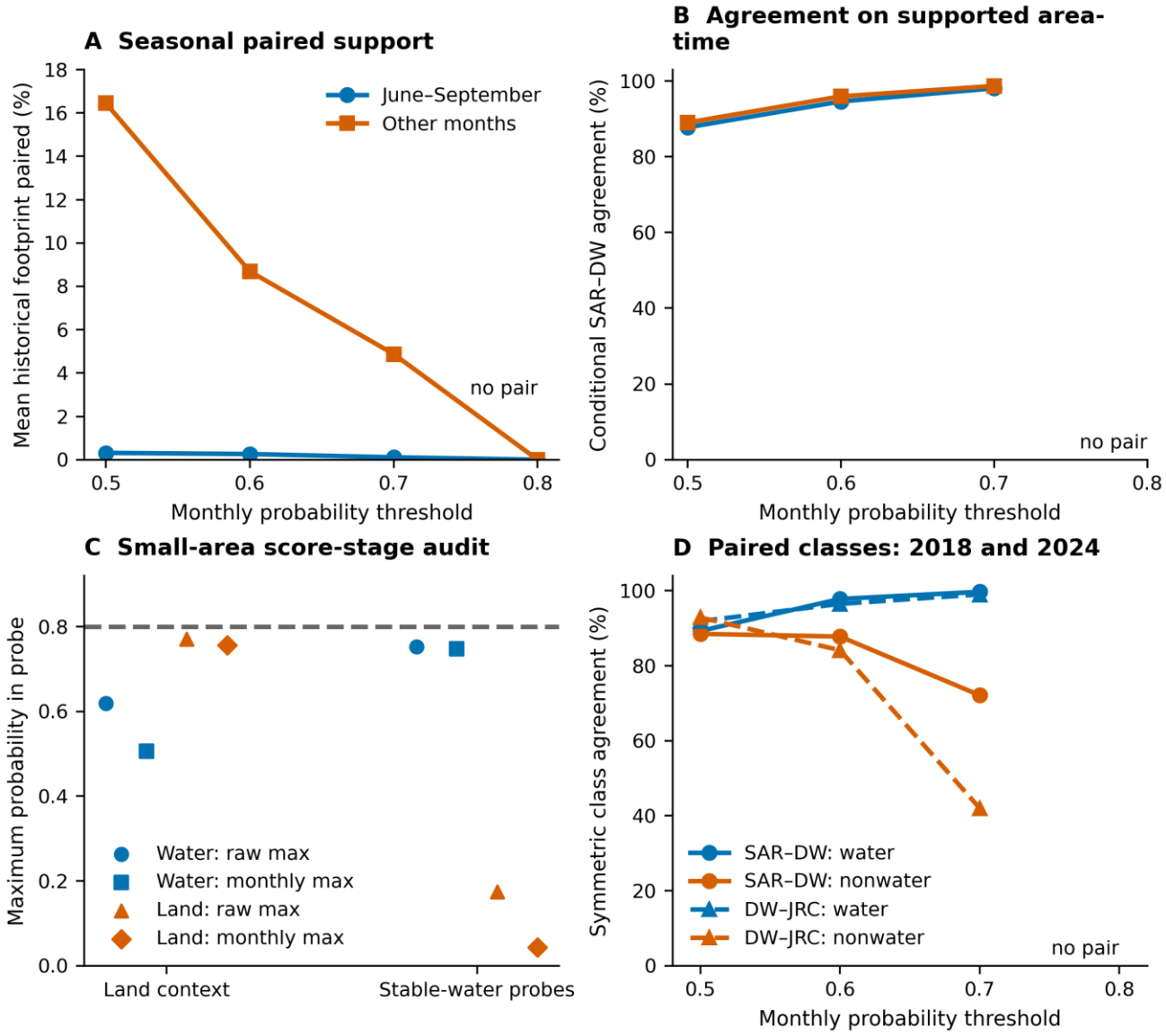


Figure 8. Seasonal and class-specific limits to satellite agreement. (A) Mean SAR–Dynamic World paired support in June–September and other months during 2015–2024, retaining zero-support months. (B) Agreement pooled by paired area-time in those seasons. (C) November 2024 maximum raw and monthly-median probabilities in one small land-context probe and three stable-source-water interiors; probes are selected diagnostics, not independently labeled validation sites. The dashed line is threshold 0.8. (D) Symmetric positive (water) and negative (nonwater) product agreement in the completed 2018/2024 union historical ever-water pilot, pooled across all 24 intended months. Product pairs have different support. Threshold 0.8 has no pairs; its class agreement is unavailable. Panels A–B cover ten years, whereas D covers the two pilot years. No panel estimates independent accuracy.

3.4 Assisted dated evidence leaves a substantial unresolved comparison frame

The 720 sampled pixel-months encompassed 3,022 individual scene-point views across 69 source Sentinel-2 scenes. Initial spectral triage assigned 337 automatic proxy labels, routed 245 complex pixel-months to date-by-date visual review, and identified 138 pixel-months with insufficient clear-sky support. Across the 245 human-reviewed pixel-months, 694 date-specific evaluations were recorded (430 water, 236 nonwater, and 28 uncertain). Aggregating dated evaluations within each pixel-month yielded 194 consistently binary human-reviewed points, whereas 35 points exhibited within-month hydrological change and 16 remained uncertain due to conflicting or ambiguous spectral signals (Figure 9; Table 3).

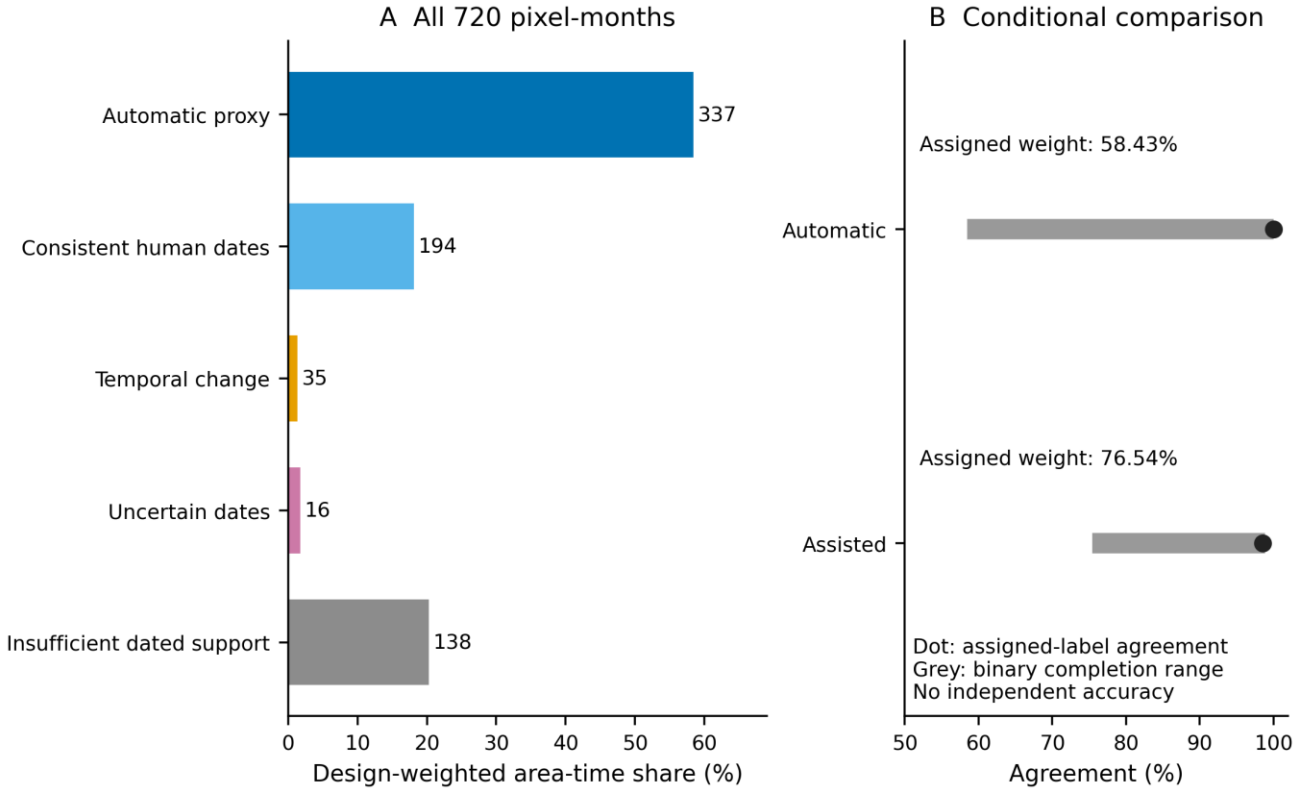


Figure 9. Dated proxy evidence and unknown comparisons for all 720 stratified pixel-month samples. (A) Design-weighted outcome shares; annotations give sample counts. (B) Dots indicate conditional agreement on assigned labels; grey segments show algebraic ranges under arbitrary binary completion of unresolved comparisons. These are not physical-accuracy bounds. Repeated scene dates are within-sample evidence, and selected human review was assisted.

Table 3. Outcomes and comparison accounting for all 720 sampled pixel-months. In panel A, weight is the share of all sampled area-time for each outcome; in panel B, it is the assigned share for each comparison route. Agreement uses assigned weight only. Completion ranges arbitrarily assign unresolved comparisons to agreement or disagreement; they are neither confidence intervals nor physical-water accuracy bounds. The assisted conditional sampling interval is 96.85–99.87%; the automatic conditional bootstrap is degenerate and omitted. Dashes denote inapplicable quantities.

Outcome or comparison route	Pixel-months (n)	Weight (%)	Agreement (%)	Completion range (%)
A. Outcomes for all sampled pixel-months				
Automatic proxy assigned	337	58.43	—	—
Consistent human dates assigned	194	18.11	—	—
Temporal change: unresolved	35	1.38	—	—
Uncertain dates: unresolved	16	1.80	—	—
Insufficient dated support: unresolved	138	20.28	—	—
Total	720	100.00	—	—
B. Assigned comparison routes				
Automatic only	337	58.43	100.00	58.43–100.00
Assisted proxy	531	76.54	98.56	75.44–98.90

Combining automatic spectral proxies (337 points; 58.43% of design-weighted area-time) with consistent human-reviewed dates (194 points; 18.11% of design weight) resolved 531 pixel-months, representing 76.54% of the total design-weighted area-time (leaving 189 pixel-months or 23.46% of design weight unresolved). Within the resolved subset, assisted conditional agreement with JRC monthly classifications was 98.56% (95% stratified bootstrap interval: 96.85–99.87%). Accounting for the 23.46% unresolved design weight without imputation yields an algebraic binary-completion range of 75.44–98.90% across the entire four-month sampling frame (Table 3).

Within the automatic-only subset (58.43% of design weight), conditional agreement with JRC was 100.00% (yielding an unassisted binary-completion range of 58.43–100.00%; Table 3). Incorporating multi-date human review expanded the resolved area-time share by 18.11 percentage points while maintaining high cross-product consistency. Because both the spectral triage and Dynamic World rely on Sentinel-2 optical acquisitions, these metrics quantify multi-resolution optical consistency across clear-sky observations rather than independent ground-truth accuracy.

4 Discussion

4.1 Seasonal decline is a monitoring signal with a defined spatial scope

The mapped seasonal decline is supported by more than the choice of two endpoint years, but its magnitude depends on the comparison. The 2000–2021 endpoint decrease was 25.42%, whereas five-year means gave 14.70% on both the primary and 2000–2021 complete footprints (C1 and C3) and 18.79% on the baseline-water cohort (C4). Whole-series associations and these window comparisons support investigating seasonal source-class contraction within retained areas. They do not justify one district-wide loss percentage. District uncertainty also matters: the controlled seasonal models retained clearer evidence in Naogaon and Rajshahi than in Chapainawabganj.

Several mechanisms remain compatible with the observed seasonal-class loss. Water occurrence may have changed, pixels may have moved between seasonal and permanent classes, or the interaction of observation timing and processing may have affected annual assignment. The fixed footprint removes changes in which pixels enter each annual area sum, but it does not remove class error or uneven seasonal sampling. Neither a persistent time coefficient nor a climate-adjusted residual identifies irrigation abstraction, land conversion, river management or an ecological mechanism.

Static landscape context narrows where follow-up could begin. The larger proportional seasonal decrease away from mapped rivers in 2020 cropland was consistent across the tested buffers, while the distant noncropland group gained seasonal-class area. However, a 2020 land-cover label cannot describe cultivation throughout 2000–2021, and a global centerline buffer cannot separate all river water from ponds. River migration, wetland conversion, crop-water management and observation effects remain rival explanations. No local groundwater or discharge time series was acquired to discriminate among them.

Regional evidence of rainfall heterogeneity (Esha et al., 2026) and coupled climatic and groundwater stress (Sarker et al., 2026) makes those mechanisms relevant hypotheses. Testing them would require matched observations of groundwater levels, abstraction, river regulation, land use and waterbody management. Basin-scale separation of storage components uses different evidence, including process models and gravity observations (Prusevich et al., 2025). Our classified areas do not supply equivalent information about water volume. Likewise, the surface-water expansion reported in southwestern Bangladesh (Huda et al., 2021) is compatible with a different regional trajectory; neither result should be generalized across the country without harmonizing domains and definitions.

The permanent class illustrates the risk of interpreting endpoints as a robust process trend. Its 41.04% endpoint increase is a valid description of source classifications on the retained footprint. However, the annual time coefficients lost clear positive support after the 2016 indicator. The appropriate interpretation retains both findings. A continuous increase in physically permanent water, an increase in storage, and improved dry-season water security are not established by this record.

4.2 Agreement should be evaluated together with the area it represents

Raising the Dynamic World threshold removed much of the classification and paired footprint while increasing agreement. This is consistent with selecting clearer, more homogeneous cases, although the present analysis does not isolate the causes of selection. High agreement on that subset cannot describe the omitted mixed, cloudy or weak-probability areas. Threshold 0.8 is particularly instructive: an apparently demanding confidence rule produced no comparison support for this implementation and domain. Treating its missing results as dry land would reverse their meaning.

The completed joint cells identify a further selection effect: the high-threshold paired population was almost entirely classified as water. Thus, strong pooled agreement need not imply comparable nonwater agreement. The two-year pilot separated 99.72% water from 72.14% nonwater SAR–Dynamic World agreement, and 99.01% from 42.02% for Dynamic World–JRC. These are symmetric comparisons between products on changing paired populations, not reference-based class accuracies. Their interpretation depends on the retained area, class prevalence and sparse seasonal support. They do not establish the corresponding class-specific ratios in every year of 2015–2024.

The assisted sample has a related support limitation. The hybrid comparison expanded assigned support beyond the highly selected automatic subset, while some interpretations remained unresolved. Reporting 76.54% assigned area-time alongside 98.56% conditional agreement makes the coverage limitation visible. The binary-completion range additionally shows the influence of unresolved weight without assuming random missingness or inventing labels. It still depends on the assigned proxy labels and cannot account for their possible errors. Accuracy-assessment literature requires reference quality and sample design to be evaluated separately (Olofsson et al., 2014; Stehman and Foody, 2019). Our assisted route meets a different purpose: documenting available evidence and unresolved support.

Cross-sensor studies with external reference imagery (Vanderhoof et al., 2023) and monitoring studies incorporating field observations (Slinski et al., 2025) provide stronger designs for future validation. Their reported performance cannot be transferred to the thresholds or wetland settings used here. The Bangladesh flood-insurance comparison similarly concerns a different outcome, but its finding that data choice affects decision performance (Saunders et al., 2025) supports the practical value of retaining divergence and support information. For district monitoring, agreement should accompany explicit maps or summaries of the population actually compared.

Independent reference datasets remain critical for long-term satellite validation. Recent initiatives to generate high-resolution hand-labeled surface-water validation datasets globally (e.g., Mukherjee et al., 2024) demonstrate the expanding potential for independent optical benchmarking. Integrating such high-resolution reference products across the Barind tract represents a valuable direction for future research, particularly to calibrate sub-pixel seasonal water transitions against moderate-resolution satellite archives.

4.3 Climate associations and observation continuity in drought monitoring

Climate-adjusted modeling clarified the association between meteorological drivers and seasonal water dynamics. The parsimonious controlled model—incorporating monsoon rainfall, observation effort, and the 2016 sensor update—demonstrated strong predictive stability (RMSE = 0.0486) while accounting for key observational shifts. While meteorological variations explain a portion of interannual water fluctuations, the consistent downward trend in seasonal water across Naogaon and Rajshahi suggests that broader hydrological and land-use factors warrant localized field monitoring.

Evaluating hydrological drought resilience requires continuous observation across seasonal cycles. When meteorological droughts occur, tracking the response of local surface water bodies depends heavily on satellite revisit frequency and cloud-free coverage. Our analysis revealed that cloud cover during monsoon-to-dry season transitions frequently interrupts continuous monitoring. Establishing dedicated sentinel monitoring networks combining automated gauging stations with high-frequency satellite observations will be essential for tracking post-drought surface water dynamics across the Barind tract.

Additional observations should be targeted to seasonal transition periods, especially pre-monsoon low-flow conditions and post-monsoon recession months. A future operational monitoring framework would benefit from a predefined network of sentinel waterbodies with repeated dated imagery and in-situ water-level observations, coupled to a stable multi-sensor reference method.

4.4 Implications for sustainable water management

For the target districts, the results support a staged use of satellite evidence. Seasonal-class losses on the retained footprint can identify areas for field or higher-resolution investigation. Before using those areas to prioritize storage restoration or water allocation, managers would need to determine whether the signal reflects persistent inundation change, land use, channel movement or classification behavior. The present district aggregates do not locate a validated restoration site or quantify a water-supply benefit.

A practical district water-monitoring dashboard should integrate three core metrics: classified surface-water area, observation frequency across key agricultural months, and the spatial fraction of the domain with verified multi-sensor agreement. Reporting these indicators concurrently preserves the critical distinction between nominal data availability and genuine observation continuity.

This approach is relevant to monitoring the extent of water-related ecosystems and to climate-adaptation planning. It is not a complete official assessment of Sustainable Development Goal indicator 6.6.1, which has its own ecosystem definitions, reporting dimensions and national implementation requirements.[3] Water quality, groundwater storage, biodiversity outcomes, household water access and the effects of management interventions were not measured. The management contribution is therefore better-specified evidence for investigation and monitoring, without claiming observed improvements in water security.

4.5 Limitations and transferability

Spatial domain definition is a primary consideration when interpreting long-term satellite trend estimates. Requiring complete annual observation validity ensures that identical pixel populations are compared across years, avoiding compositional shifts caused by time-varying missing data. While the 2000–2024 reference footprint (C1) incorporates post-2021 observation validity, our sensitivity framework demonstrated nearly identical five-year declines on the strictly non-retrospective 2000–2021 cohort (C3; −14.70%) and the 2000–2004 baseline-water cohort (C4; −18.79%). Nevertheless, any annually valid

[3] UNEP description and methodology for indicator 6.6.1: https://www.unep.org/en/goal-6-clean-water-and-sanitation/indicator-661.

footprint excludes transient water bodies that are systematically obscured by persistent cloud cover, meaning that retained cohorts represent consistently observable water-occurrence zones rather than a complete district-wide census. Furthermore, because only November retained at least 20% of the baseline footprint across all 22 years, monthly optical archives cannot support a fully balanced twelve-month hydroperiod trend analysis without multi-sensor integration.

Source classification and alignment remain important. Corrected annual products, Landsat collection differences, mixed boundary pixels, SAR threshold behavior, wet soil, rice and vegetation can all affect water estimates. The method envelopes cover specified thresholds and unknown support, while excluding some systematic errors. Dynamic World and the dated optical proxy have related inputs, and monthly sensor composites do not necessarily represent the same day. Independent annual seasonal/permanent accuracy remains unmeasured.

Statistical uncertainty is also incomplete. Twenty-two annual observations provide limited information for models with multiple climate, effort and processing terms. HAC intervals are approximate and pointwise. Reported adjustment families cover specified district tests, not the entire exploratory workflow; spatial dependence and unobserved drivers remain. Retrospective cohorts make the temporal checks conditional hindcasts. The four-month proxy bootstrap addresses sampled pixel-month variation, without resolving nonrandom reference failure or spatial clustering. The lack of independent labels is explicit and prevents presenting proxy agreement as accuracy.

These limitations bound transferability to similar monitoring problems. Estimates should be reported together with their spatial, temporal and reference support. The numerical trends and thresholds themselves remain specific to the declared products, domains and periods in northwestern Bangladesh.

5 Conclusion

Mapped seasonal JRC source-class area decreased within the retained annual-observation footprint in northwestern Bangladesh. The 2000–2021 endpoint decline was 25.42%, compared with 14.70% between five-year means on the same footprint and 18.79% on a baseline-water complete cohort. Negative seasonal associations persisted across the tested specifications, with stronger controlled district evidence in Naogaon and Rajshahi. Permanent-class endpoint gains remained sensitive to observation and processing controls. These findings concern classified areas and declared spatial support, not district-wide water storage.

These complementary analyses underscore the importance of observational support in remote-sensing-based water assessment. Cross-sensor evaluation demonstrated that high radar–optical agreement is largely confined to permanent water bodies, while optical-radar divergence remains pronounced in seasonal and vegetated margins. Furthermore, dated sample validation demonstrated high classification consistency in clear acquisitions while highlighting persistent seasonal cloud constraints. Satellite-based monitoring in drought-prone regions must therefore explicitly account for observation frequency and sensor support alongside mapped changes. These findings establish robust baseline evidence to guide field investigation and targeted water resource planning across northwestern Bangladesh.

Data and code availability

The code, processed data, and analysis pipelines supporting the reported calculations are publicly available in the project repository at https://github.com/asifahamed11/northwestern-bangladesh-surface-water. Complete reproduction packages—including exported observations, dated interpretation records, tabular results, analysis scripts, source manifests, and computational instructions—are also provided in the Supplementary Materials (Supplementary_Code_and_Tables.zip, Supplementary_Grid.zip, and Supplementary_Context.zip), enabling independent replication of all reported findings without proprietary dependencies.

Author contributions

Asif Ahamed: Conceptualization, Methodology, Software, Formal analysis, Investigation, Writing – original draft, Visualization.

Ahammad Hossain: Conceptualization, Formal analysis, Supervision, Writing – review & editing.

Md. Tanvir Hasan: Data curation, Software, Formal analysis, Investigation.

Most. Alisa Tabassum: Investigation, Validation, Data curation.

A.H.M. Rahmatullah Imon: Conceptualization, Supervision, Project administration, Writing – review & editing. All authors read, reviewed, and approved the final manuscript.

Conflict of interest

The authors declare that the research was conducted in the absence of any commercial or financial relationships that could be construed as a potential conflict of interest.

Funding

The authors report no funding for this research.

Acknowledgments and generative AI use

We gratefully acknowledge the organizations providing the satellite, climate, and administrative boundary products utilized in this study, including Google Earth Engine, the European Commission's Joint Research Centre, the CHIRPS precipitation project, and the ECMWF ERA5-Land team. Generative AI assistance was utilized solely for language editing and style refinement during manuscript preparation. All research design, computational code, statistical modeling, figure generation, and scientific interpretations were conducted by the authors.

Supplementary material

Supplementary Methods retain compact source, grid, label-history and reproduction notes. All nine figures and three tables are in the main manuscript. The three supplementary code/data archives include the complete tabular datasets, sensitivity outputs, multi-sensor comparison CSVs, source records, and reproduction scripts.

References

Benjamini, Y., and Yekutieli, D. (2001). The control of the false discovery rate in multiple testing under dependency. Ann. Stat. 29, 1165–1188. doi: 10.1214/aos/1013699998

Brown, C. F., Brumby, S. P., Guzder-Williams, B., Birch, T., Hyde, S. B., Mazzariello, J., et al. (2022). Dynamic World, Near real-time global 10 m land use land cover mapping. Sci. Data 9, 251. doi: 10.1038/s41597-022-01307-4

Esha, M. Z., Khan, M. S. A., and Tabassum, S. (2026). Spatial-temporal analysis of annual rainfall trends and variability in the Barind Tract. Discov. Atmos. 4, 14. doi: 10.1007/s44292-026-00082-w

Funk, C., Peterson, P., Harrison, L., Saldivar, R., Landsfeld, M., Pedreros, D., et al. (2026). The Climate Hazards Center Infrared Precipitation with Stations, Version 3. Sci. Data 13, 718. doi: 10.1038/s41597-026-07096-4

Gorelick, N., Hancher, M., Dixon, M., Ilyushchenko, S., Thau, D., and Moore, R. (2017). Google Earth Engine: Planetary-scale geospatial analysis for everyone. Remote Sens. Environ. 202, 18–27. doi: 10.1016/j.rse.2017.06.031

Huda, N., Terao, T., Nonomura, A., and Suenaga, Y. (2021). Time-Series Remote Sensing Study to Detect Surface Water Seasonality and Local Water Management at Upper Reaches of Southwestern Bengal Delta from 1972 to 2020. *Sustainability* 13, 9798. doi: 10.3390/su13179798

Lehner, B., and Grill, G. (2013). Global river hydrography and network routing: baseline data and new approaches to study the world's large river systems. Hydrol. Process. 27, 2171–2186. doi: 10.1002/hyp.9740

Mukherjee, R., Policelli, F., Wang, R., Arellano-Thompson, E., Tellman, B., Sharma, P., et al. (2024). A globally sampled high-resolution hand-labeled validation dataset for evaluating surface water extent maps. Earth Syst. Sci. Data 16, 4311-4323. doi: 10.5194/essd-16-4311-2024

Muñoz-Sabater, J., Dutra, E., Agustí-Panareda, A., Albergel, C., Arduini, G., Balsamo, G., et al. (2021). ERA5-Land: a state-of-the-art global reanalysis dataset for land applications. *Earth Syst. Sci. Data* 13, 4349–4383. doi: 10.5194/essd-13-4349-2021

Newey, W. K., and West, K. D. (1987). A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix. *Econometrica* 55, 703–708. doi: 10.2307/1913610

Olofsson, P., Foody, G. M., Herold, M., Stehman, S. V., Woodcock, C. E., and Wulder, M. A. (2014). Good practices for estimating area and assessing accuracy of land change. Remote Sens. Environ. 148, 42–57. doi: 10.1016/j.rse.2014.02.015

Pekel, J.-F., Cottam, A., Gorelick, N., and Belward, A. S. (2016). High-resolution mapping of global surface water and its long-term changes. *Nature* 540, 418–422. doi: 10.1038/nature20584

Prusevich, A. A., Lammers, R. B., Grogan, D. S., Zuidema, S., Meko, D. M., Rounce, D. R., et al. (2025). Decomposing land surface total water storage in the Indus, Ganges, and Brahmaputra basins. *Front. Earth Sci.* 13, 1551218. doi: 10.3389/feart.2025.1551218

Sarker, T., Reza, S., Roy, R., Uddin, T., Alif, S. A., and Tariq, A. (2026). Geospatial mapping of heatwaves and drought: Exploring their link to vegetation, soil moisture, and groundwater in the thirsty Barind Tract, Bangladesh. Environ. Earth Sci. 85, 88. doi: 10.1007/s12665-025-12787-z

Saunders, A., Tellman, B., Benami, E., Anchukaitis, K., Hossain, S., Bennett, A., et al. (2025). Sensitivity to Data Choice for Index-Based Flood Insurance. *Earth's Future* 13, e2025EF005966. doi: 10.1029/2025ef005966

Slinski, K., Senay, G. B., Adoum, A., Shukla, S., McNally, A., Rowland, J., et al. (2025). In situ, modeled, and earth observation monitoring of surface water availability in West African rangelands. *Front. Water* 7, 1320010. doi: 10.3389/frwa.2025.1320010

Stehman, S. V., and Foody, G. M. (2019). Key issues in rigorous accuracy assessment of land cover products. Remote Sens. Environ. 231, 111199. doi: 10.1016/j.rse.2019.05.018

763 Van Loon, A. F. (2015). Hydrological drought explained. *WIREs Water* 2, 359–392. doi:
764 10.1002/wat2.1085

765 Vanderhoof, M. K., Alexander, L., Christensen, J., Solvik, K., Nieuwlandt, P., and Sagehorn, M. (2023).
766 High-frequency time series comparison of Sentinel-1 and Sentinel-2 satellites for mapping open
767 and vegetated water across the United States (2017–2021). Remote Sens. Environ. 288, 113498.
768 doi: 10.1016/j.rse.2023.113498

769 Xu, H. (2006). Modification of normalised difference water index (NDWI) to enhance open water
770 features in remotely sensed imagery. Int. J. Remote Sens. 27, 3025–3033. doi:
771 10.1080/01431160600589179

772 Zanaga, D., Van De Kerchove, R., De Keersmaecker, W., Souverijns, N., Brockmann, C., Quast, R., et al.
773 (2021). ESA WorldCover 10 m 2020 v100 [Dataset]. Zenodo. doi: 10.5281/zenodo.5571936

774